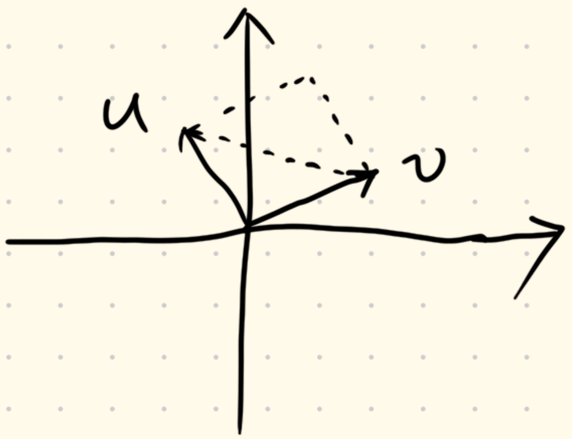
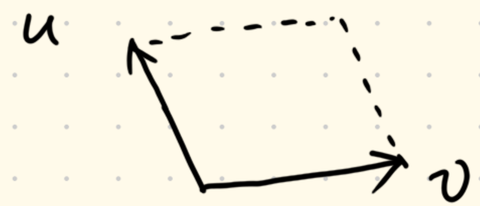
Recall: high-dimensional volume and determinant.

In $\mathbb{R}^2$, $|\det(u|v)|$.



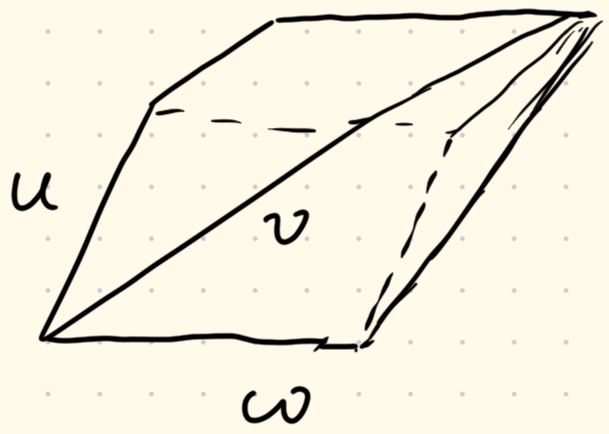
In $\mathbb{R}^3$, $\|u \times v\|$



$$\begin{pmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$$

$$= (u_2u_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1)$$

In $\mathbb{R}^3$,



$|\det(u|v|w)|$ parrallelotope.

$\frac{1}{6}|\det(u|v|w)|$ simplicial complex.

In $\mathbb{R}^n$, k dimensional parralletope spanned by $u_1, \dots, u_k$.

$A = (u_1 | \cdots | u_k)_{n\times k}$ $\sqrt{\det(A^TA)}$, proof by Gram-Schmidt orthogonalization.

$A = QR$ ↖ upper.
↑
orthogonal

prop (det) $A \in M_n(\mathbb{R})$.

1) $\det(cA) = c^n \det A$.

2) $\det(AB) = \det A \cdot \det B$.

$\det(A^k) = (\det A)^k$. $\det(A^{-1}) = (\det A)^{-1}$ if A is invertible.

$\det(P^{-1}AP) = \det A$. $\det(A^T) = \det A$.

adjoint matrix $(A^V)_{ij} = A_{ji}$.

$(\mathrm{adj}(A))_{ij}$

e.g. $A = \begin{pmatrix} 1 & 3 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{pmatrix}$ $A^\vee = \begin{pmatrix} -2 & -9 & 5 \\ 0 & 2 & -1 \\ 1 & 3 & -2 \end{pmatrix}$.

$$(A A^\vee)_{ij} = \sum_{k=1}^{n} a_{ik} A_{jk} = \delta_{ij} \cdot \det A.$$

$$A A^\vee = (\det A) I_n \quad . \quad A^{-1} = \frac{A^\vee}{\det A}.$$

$$\det(A^\vee) = (\det A)^{n-1},$$

$$(A^T)^\vee = (A^\vee)^T, \quad (cA)^\vee = c^{n-1} A^\vee.$$

$$(AB)^\vee = B^\vee A^\vee, \quad (A^\vee)^\vee = (\det A)^{n-2} A, \quad n \geq 2.$$

$$\operatorname{rank} A^\vee = \begin{cases} n, & \operatorname{rank} A = n. \ \checkmark \\ 1, & \operatorname{rank} A = n-1. \\ 0, & \operatorname{rank} A \leq n-2. \ \checkmark \end{cases}$$

$A^\vee \neq 0 \Rightarrow A \cdot A^\vee = 0$ ★

$A^\vee = 0$

$A^\vee = (v_1 | \cdots | v_n)$

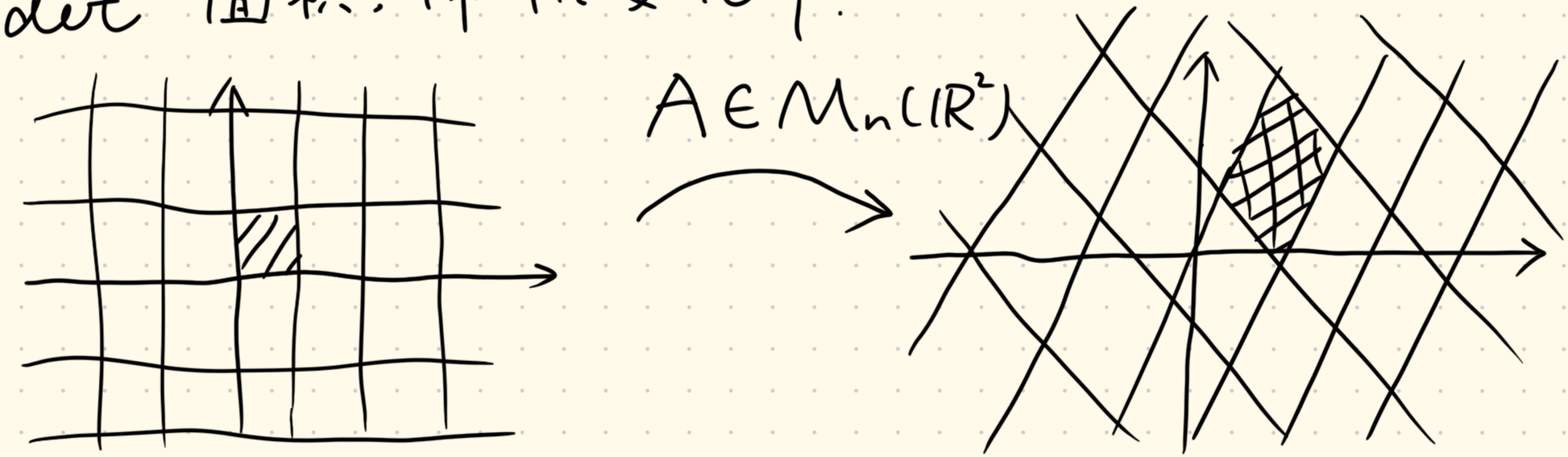
$A v_j = 0. \ \forall j$

$\dim \ker A = 1.$

Cramer's rule $Ax = b$. $x_i = \det A_i / \det A$.

where A_i is the matrix formed by replacing the i-th column of A by b.

det 面积，体积变化率.



3.3 7. $\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & 0 \\ X & I \end{pmatrix}\begin{pmatrix} A & B \\ 0 & Y \end{pmatrix} = \begin{pmatrix} A & B \\ XA & XB+Y \end{pmatrix}$.

$\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \cdot \det(D - CA^{-1}B)$

$AC = CA$

$X = CA^{-1}$, $Y = D - CA^{-1}B$.

$\det(AD - CB)$.

3.3 8. $X \in M_{n,k}(\mathbb{R})$, $Y \in M_{k,n}(\mathbb{R})$.

$M = \begin{pmatrix} I_n & X \\ -Y & I_k \end{pmatrix}$ $\det M = \det(I_k + YX) = \det(I_n + XY)$.

3.3 13.(1) $$M = \begin{pmatrix} a & b & c & d \\ -b & a & d & -c \\ -c & -d & a & b \\ -d & c & -b & a \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & b & c & d \\ -b & 0 & d & -c \\ -c & -d & 0 & b \\ -d & c & -b & 0 \end{pmatrix}}_{=K} + aI_4.$$

$$(\det M)^2 = \det(M^T M) = \det\left((aI_4 - K)(aI_4 + K)\right)$$

$$= \det(a^2 I_4 - K^2) = \det\left(a^2 I_4 + (b^2 + c^2 + d^2) I_4\right)$$

$$= \det\left((a^2 + b^2 + c^2 + d^4) I_4\right) = (a^2 + b^2 + c^2 + d^2)^4.$$

3.3 (2)(2).

$$D = \begin{vmatrix} I + a_1 + b_1 & a_1 + b_2 & \cdots & a_1 + b_n \\ a_2 + b_1 & I + a_2 + b_2 & \cdots & a_2 + b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n + b_1 & a_n + b_2 & \cdots & I + a_n + b_n \end{vmatrix}$$

$$a = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

$$D = I_n + a\mathbb{1}^T + \mathbb{1}b^T = I_n + (a \mid \mathbb{1}) \begin{pmatrix} \mathbb{1}^T \\ b^T \end{pmatrix}$$

$$\det(I_n + UV) = \det(I_2 + VU) \qquad \begin{matrix} U \\ \| \end{matrix} \qquad \begin{matrix} V \\ \| \end{matrix}$$

$$= \left(1 + \sum a_i\right)\left(1 + \sum b_i\right) - n \sum a_i b_i. \qquad VU = \begin{pmatrix} \sum a_i & n \\ \sum a_i b_i & \sum b_i \end{pmatrix}$$